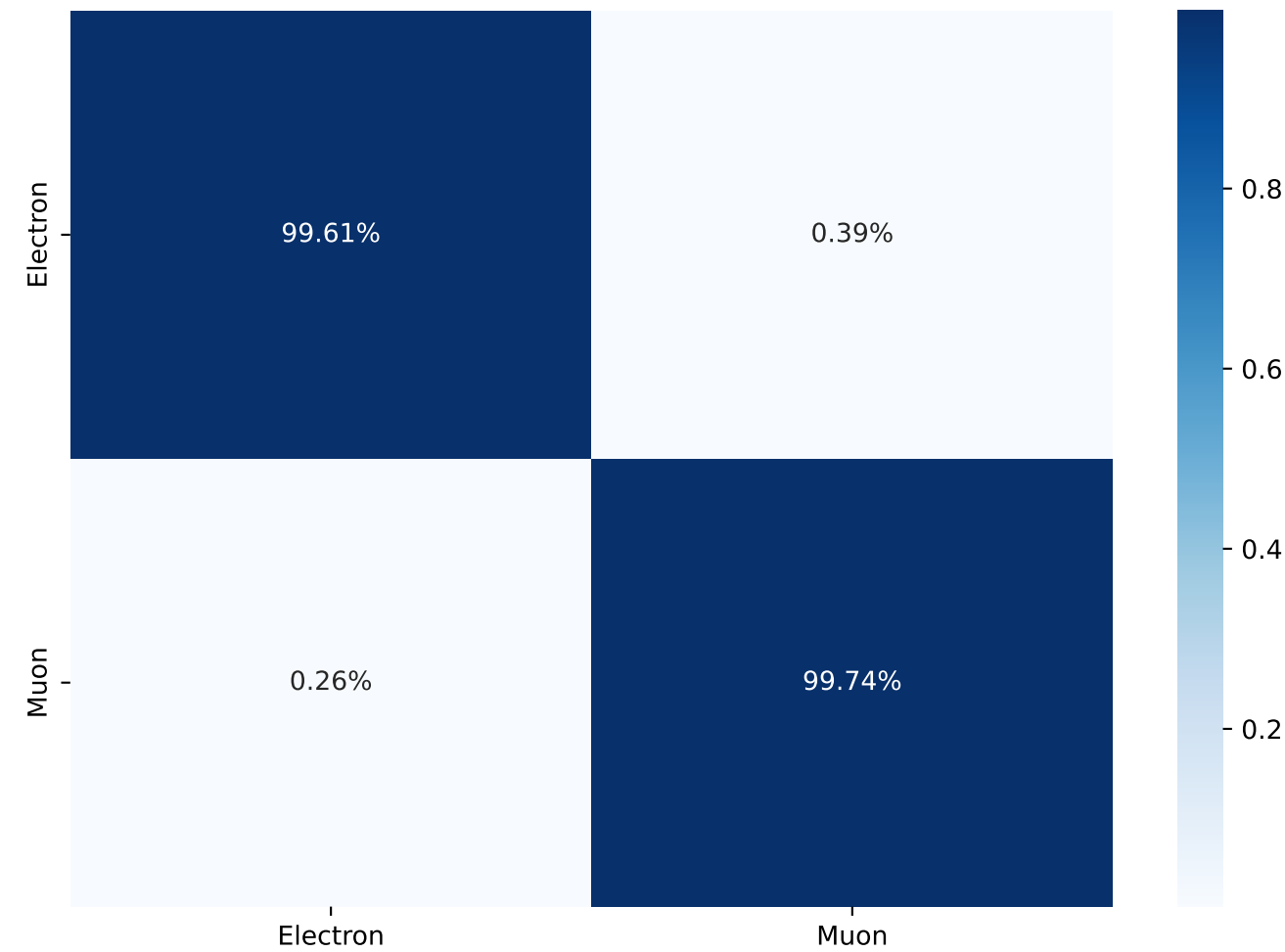
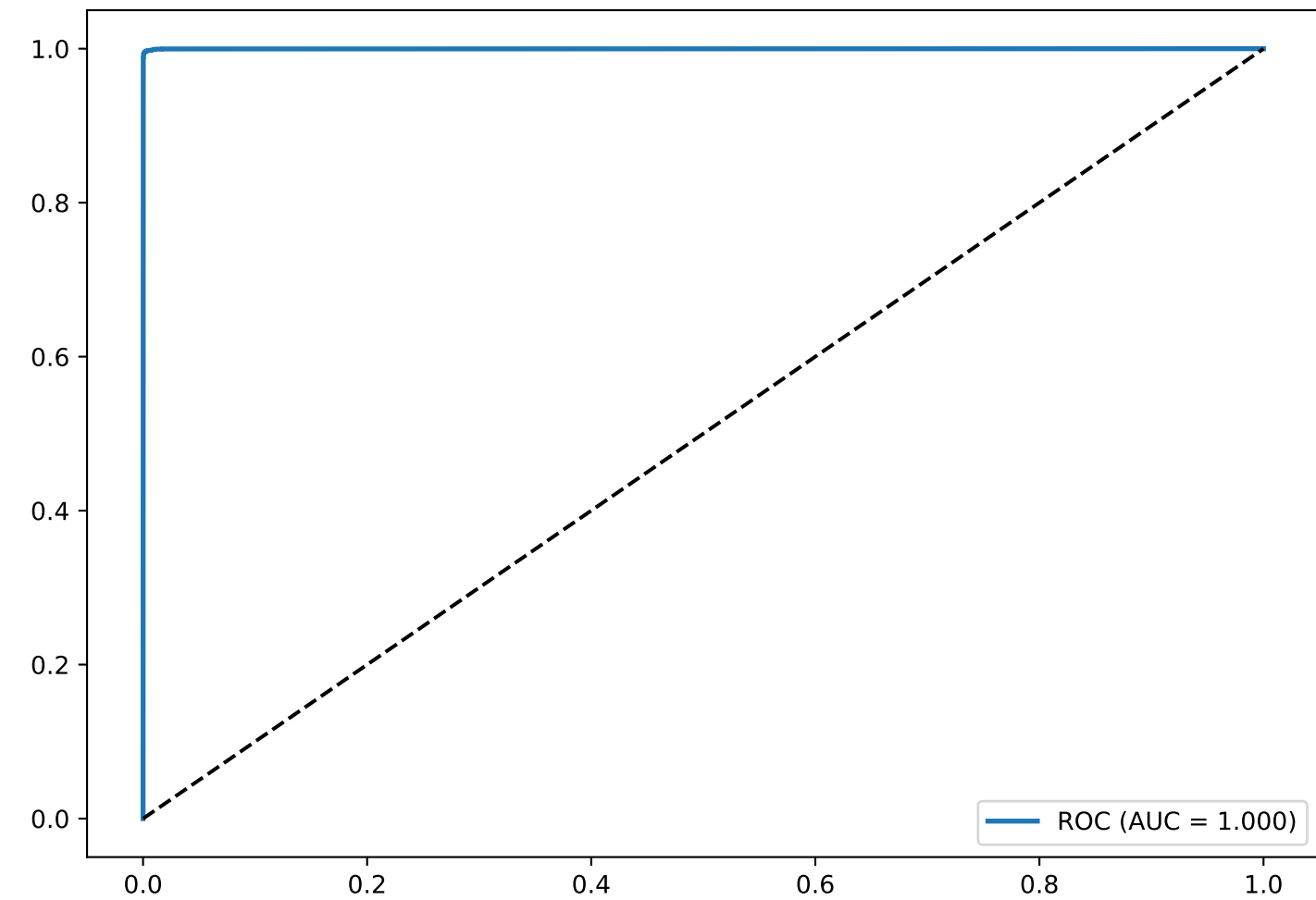


EtaRange = FullRange (TEST DATA) - Accuracy: 0.997

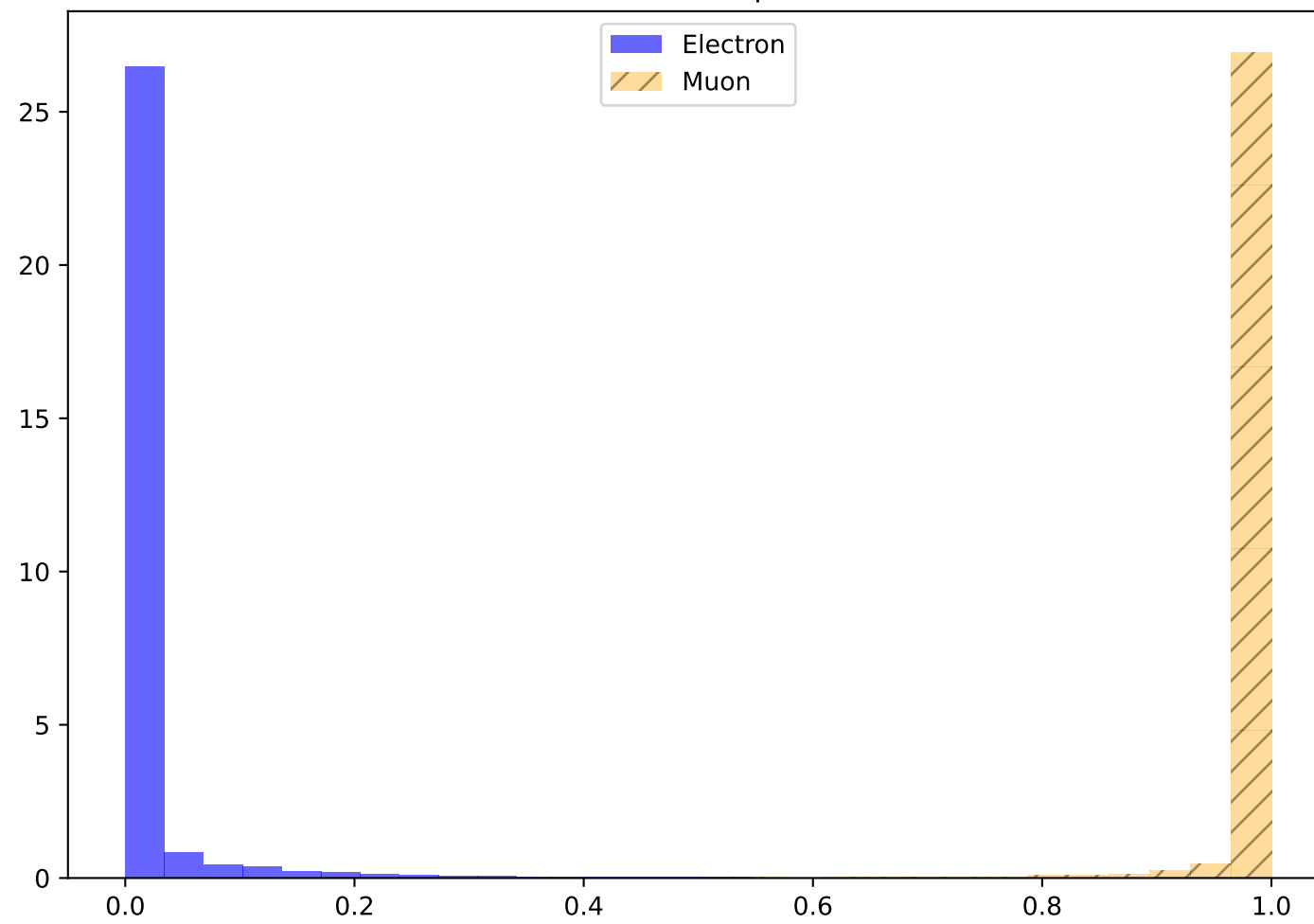
Confusion Matrix



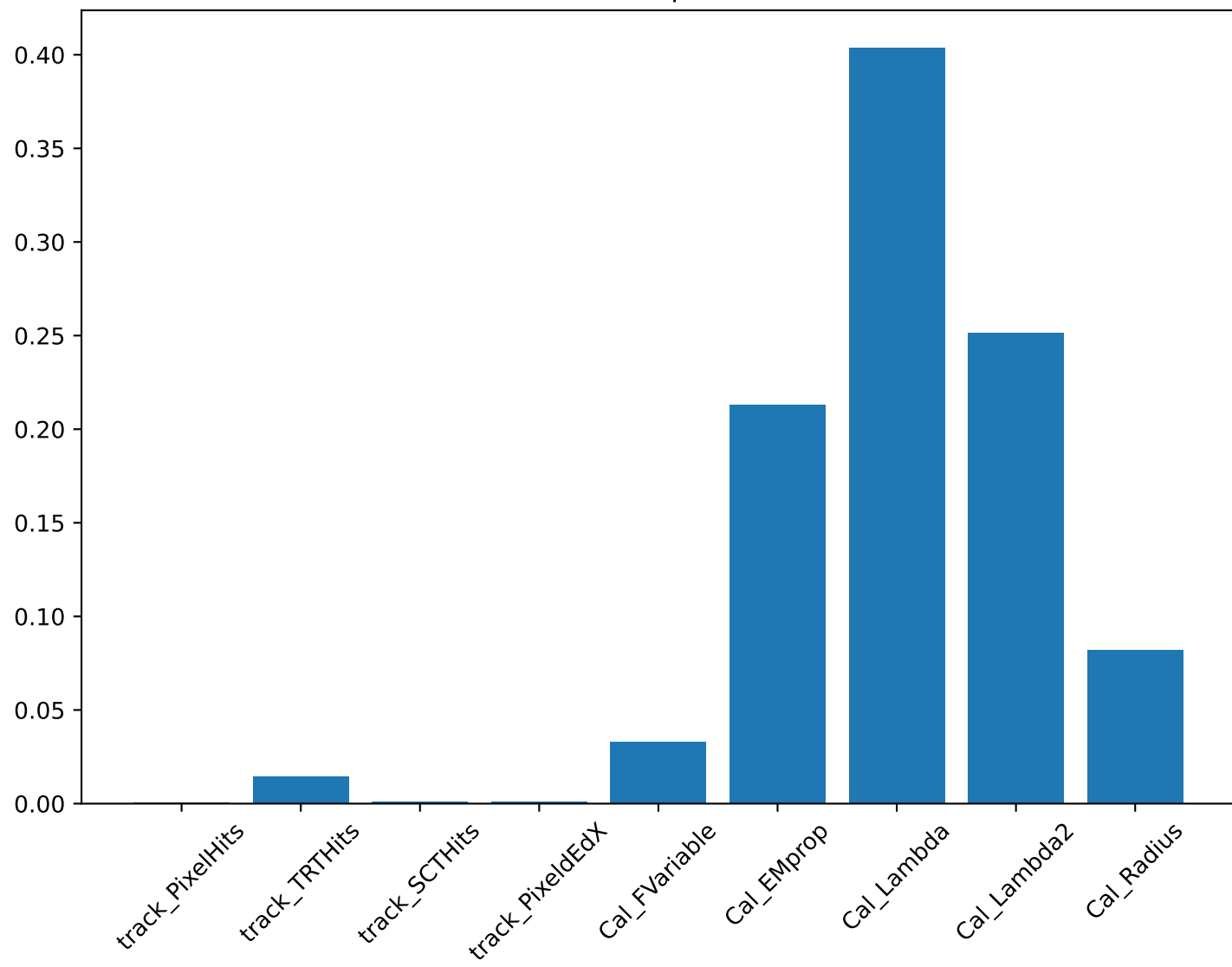
ROC Curve



Classifier Response

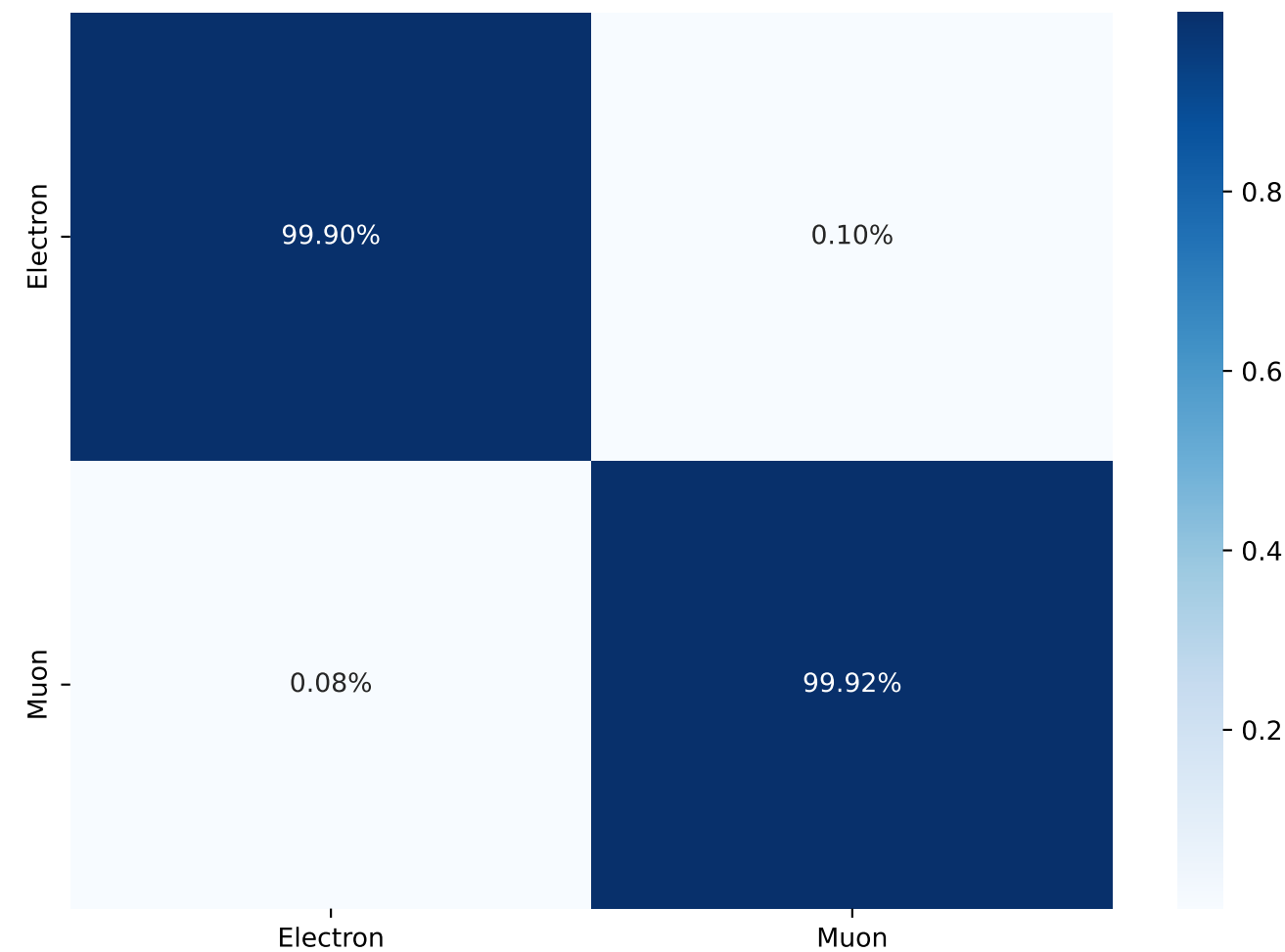


Feature Importance

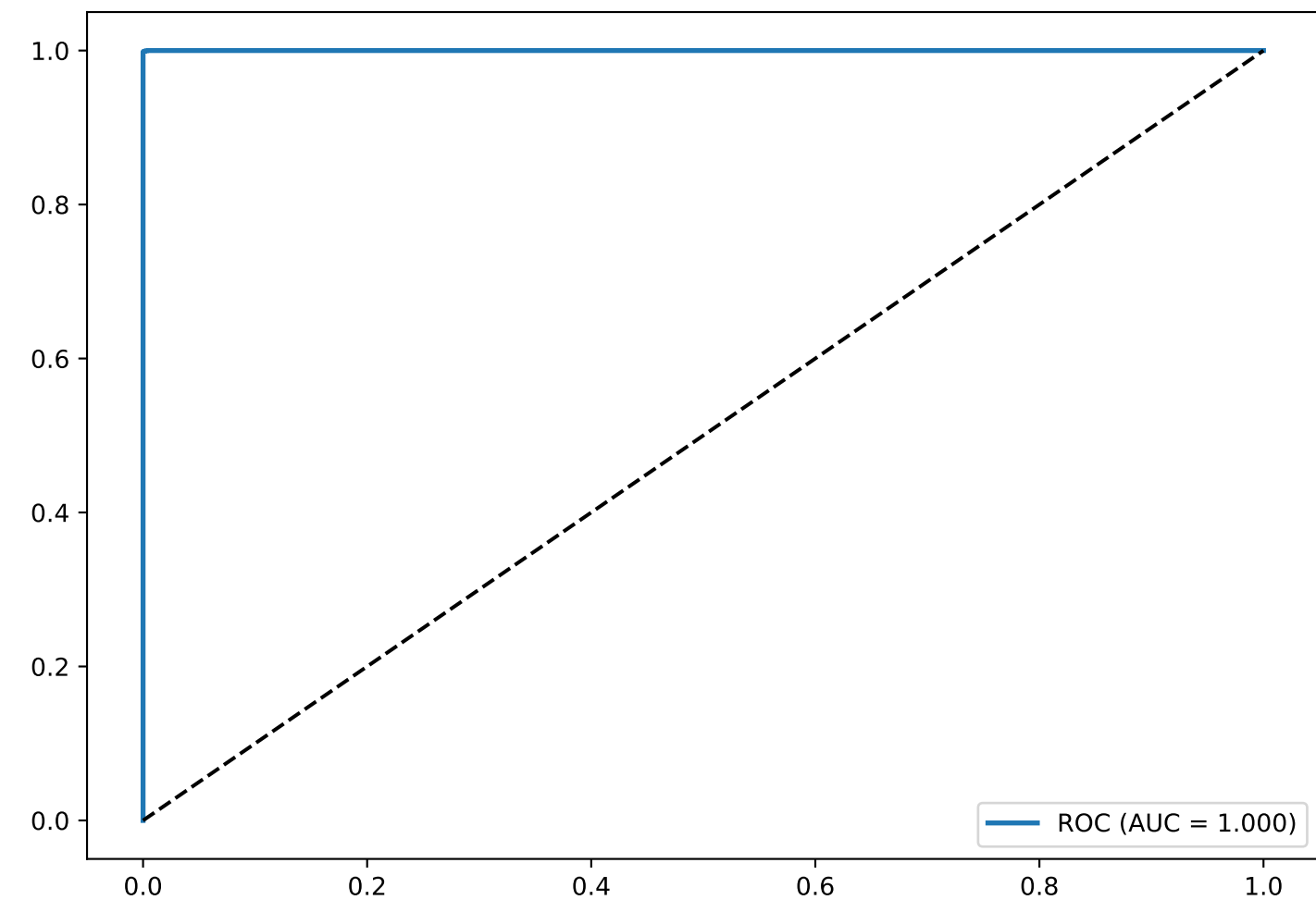


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.999

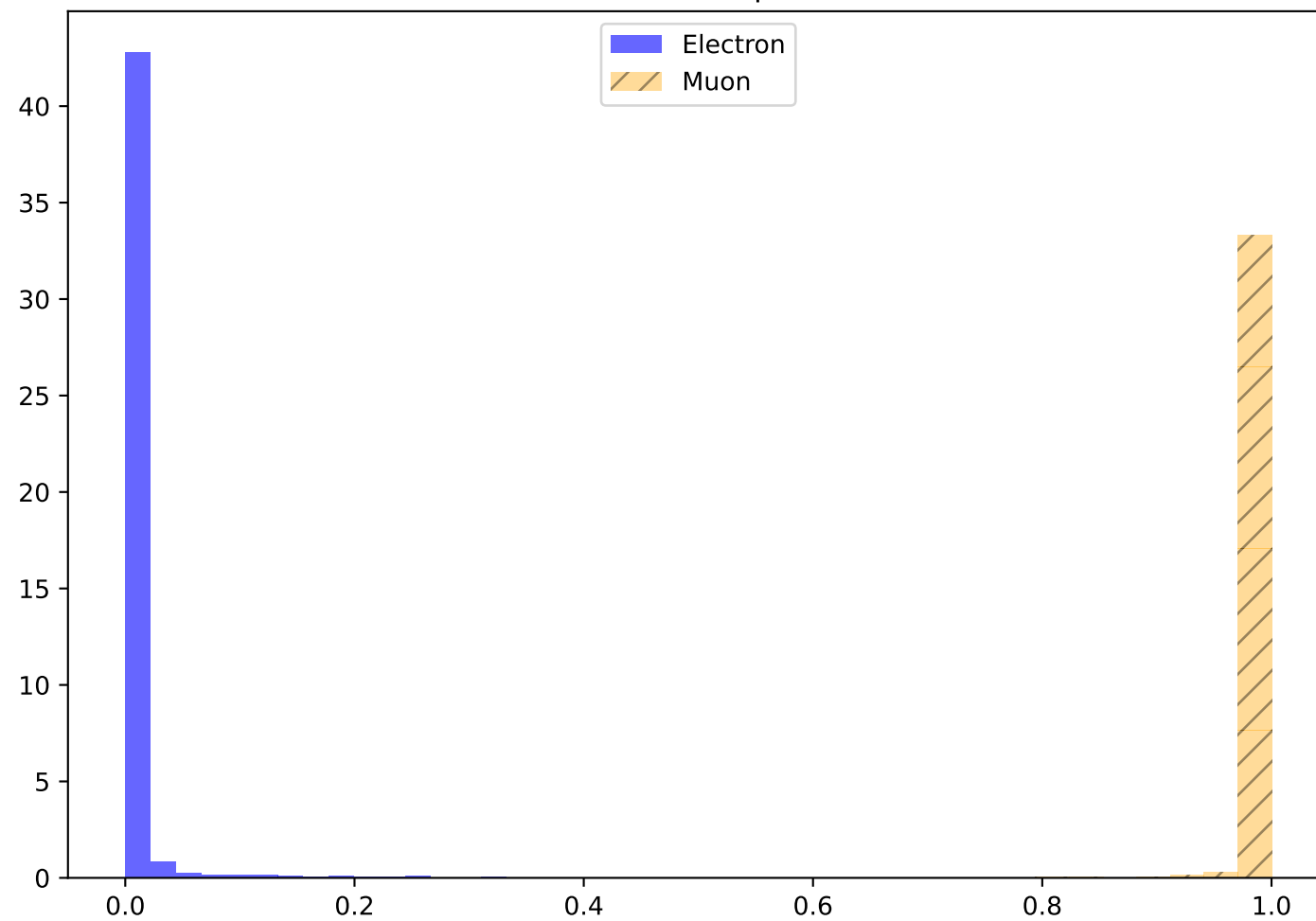
Confusion Matrix



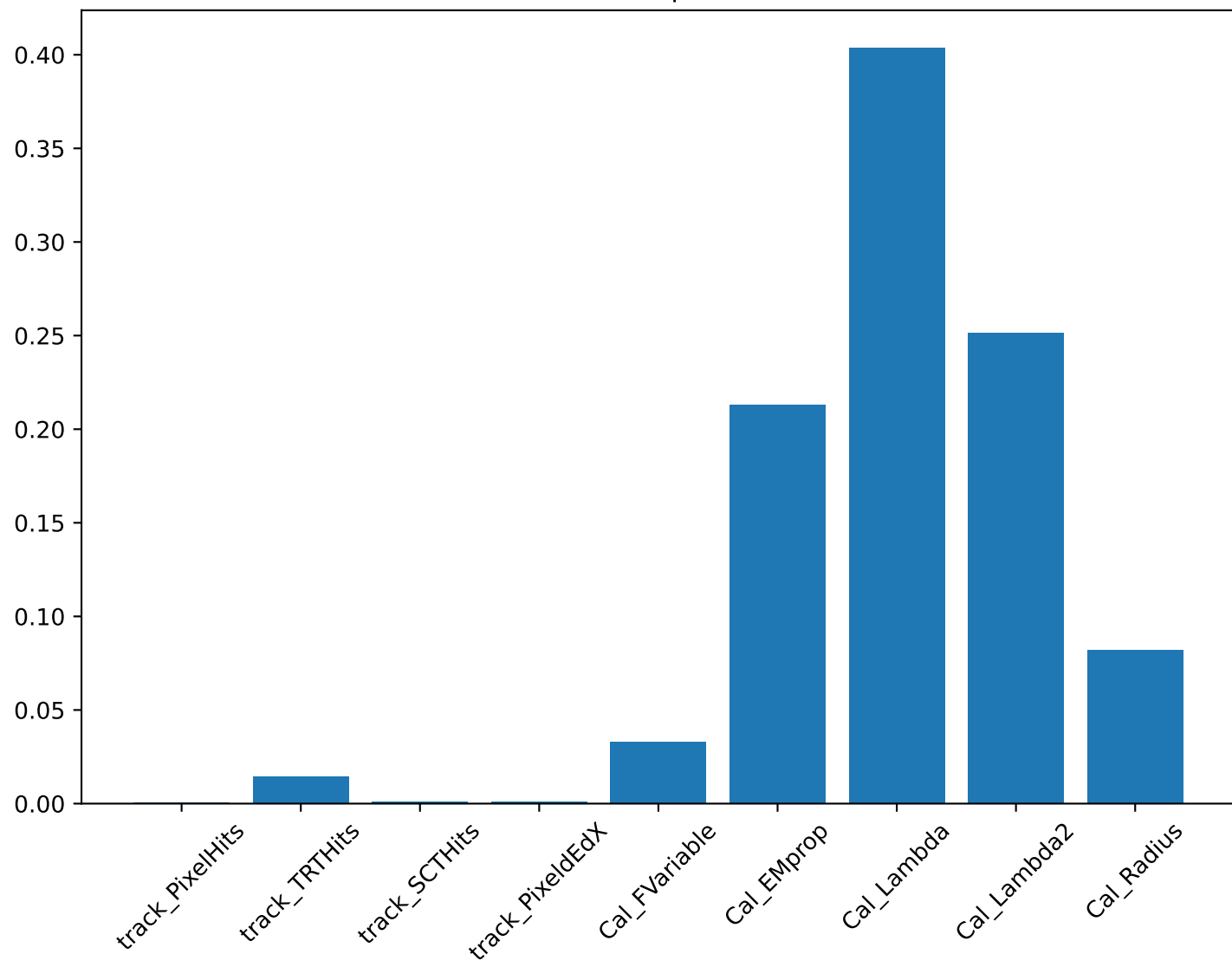
ROC Curve



Classifier Response

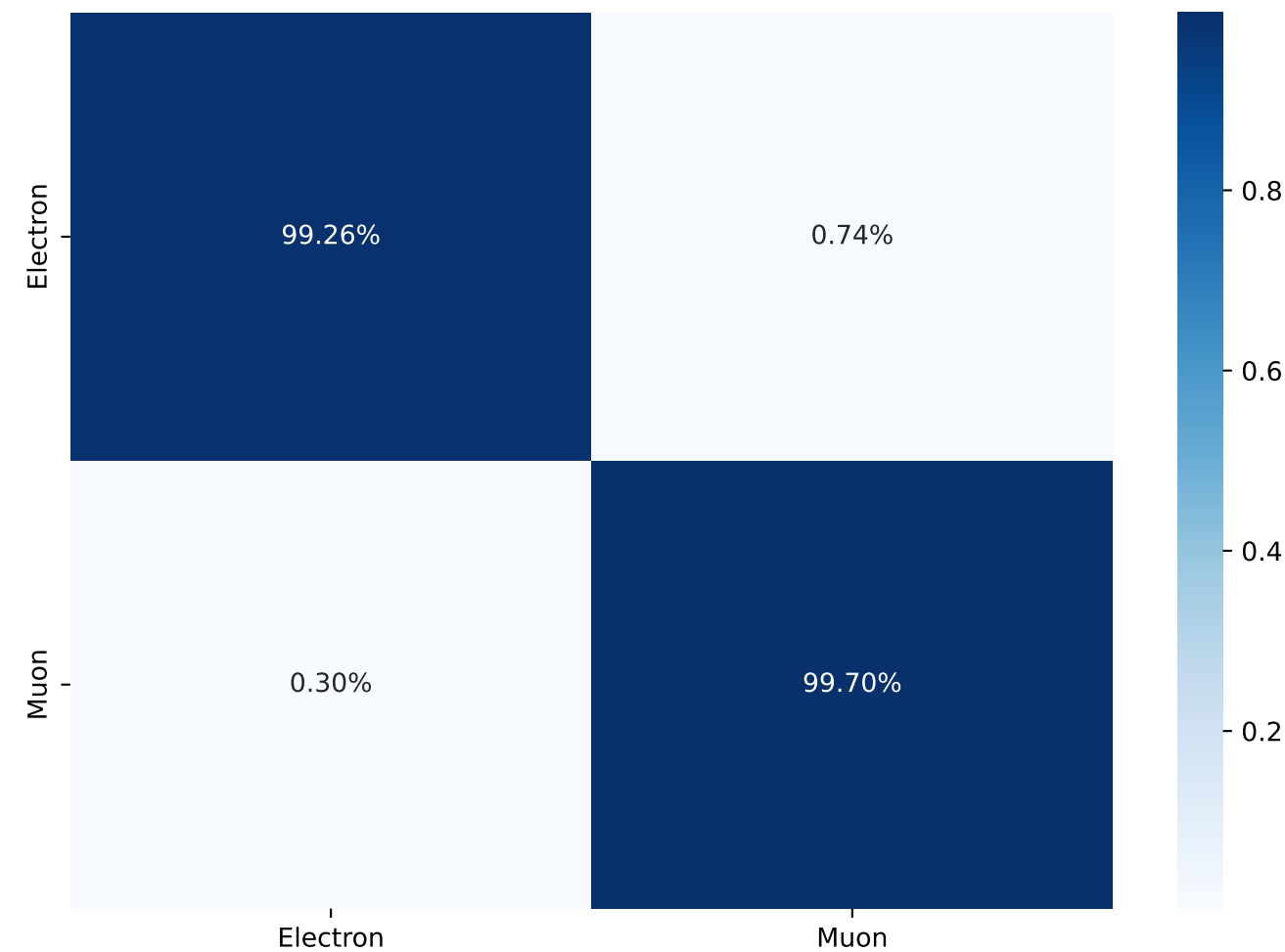


Feature Importance

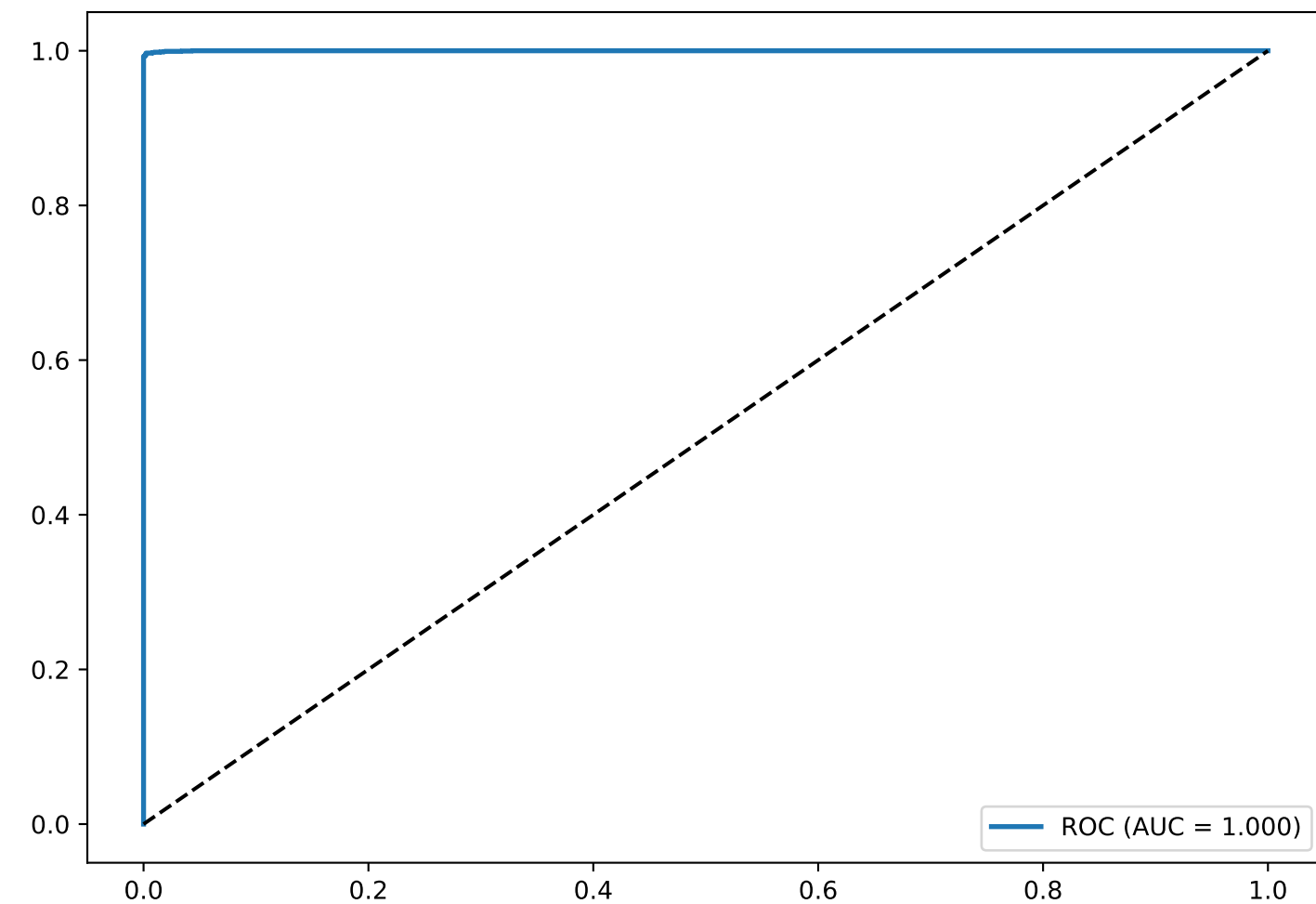


EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.996

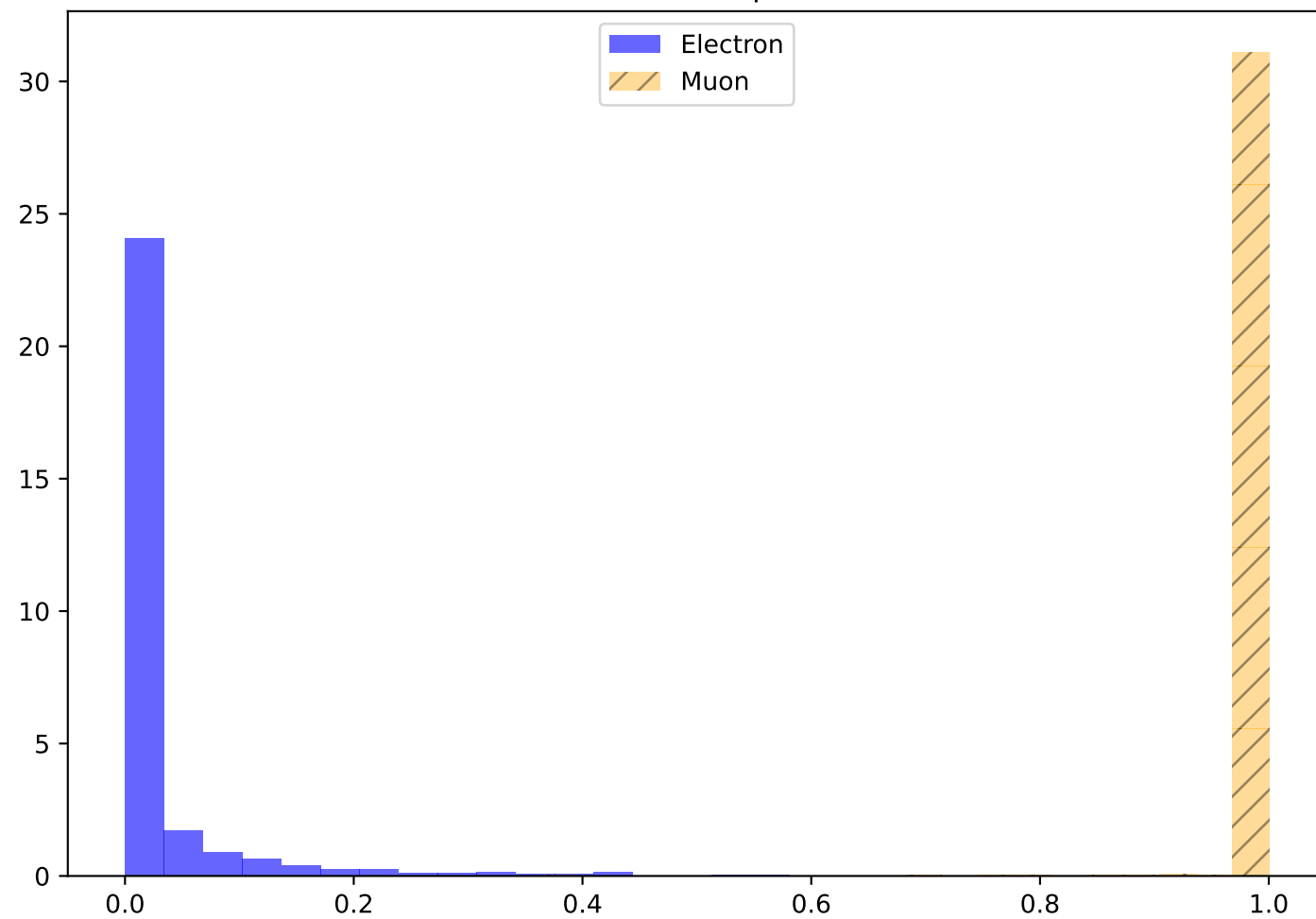
Confusion Matrix



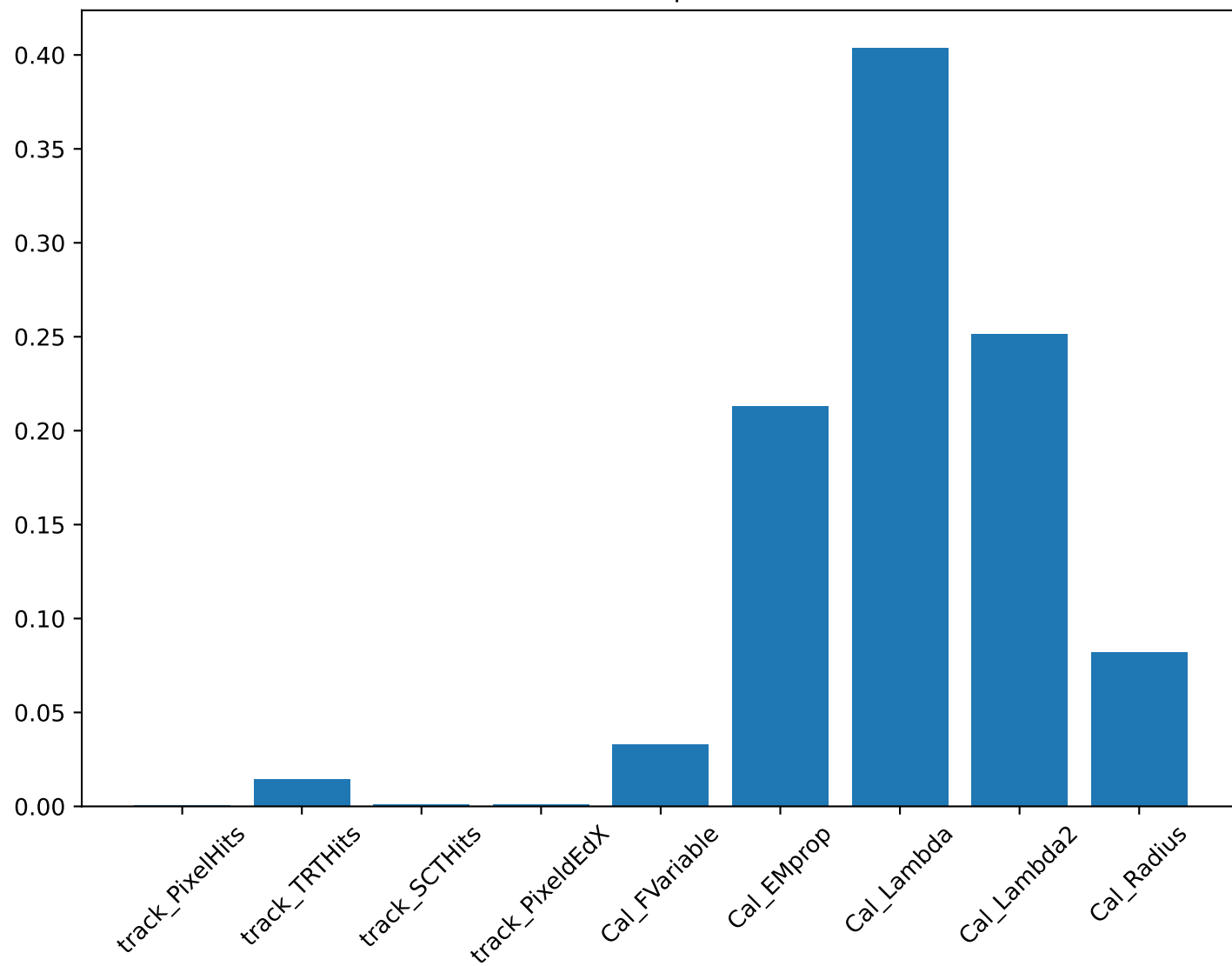
ROC Curve



Classifier Response

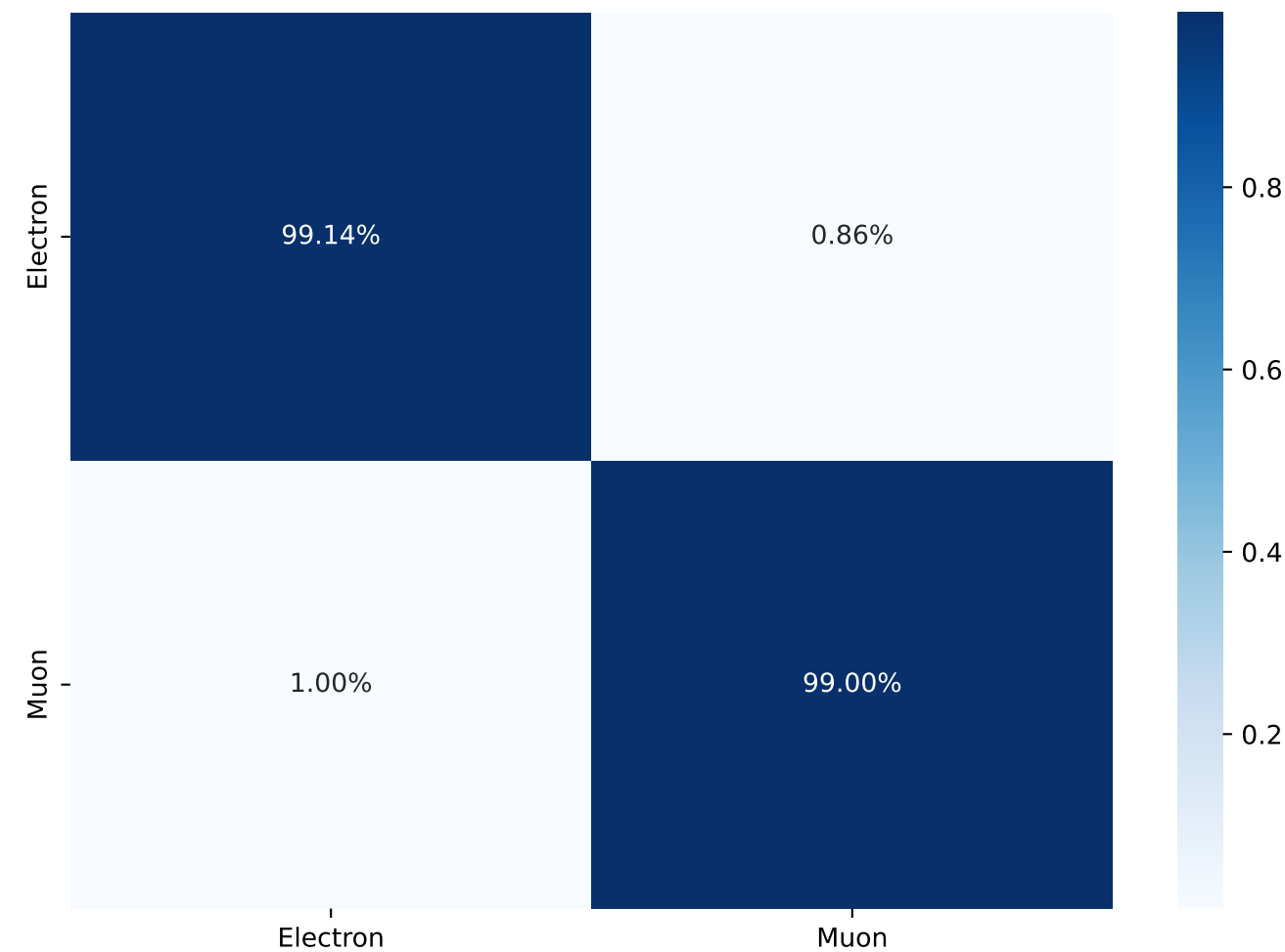


Feature Importance

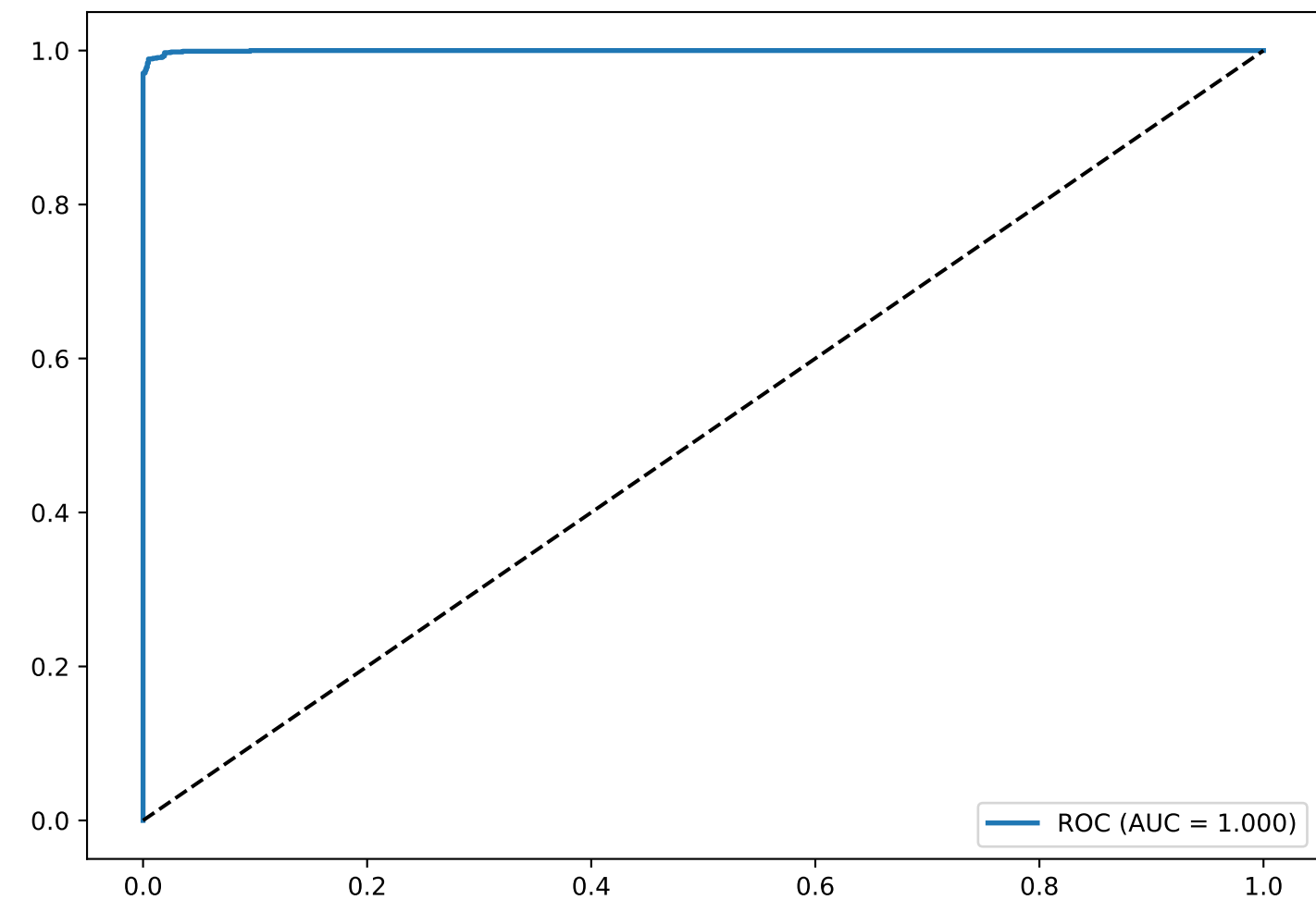


EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.991

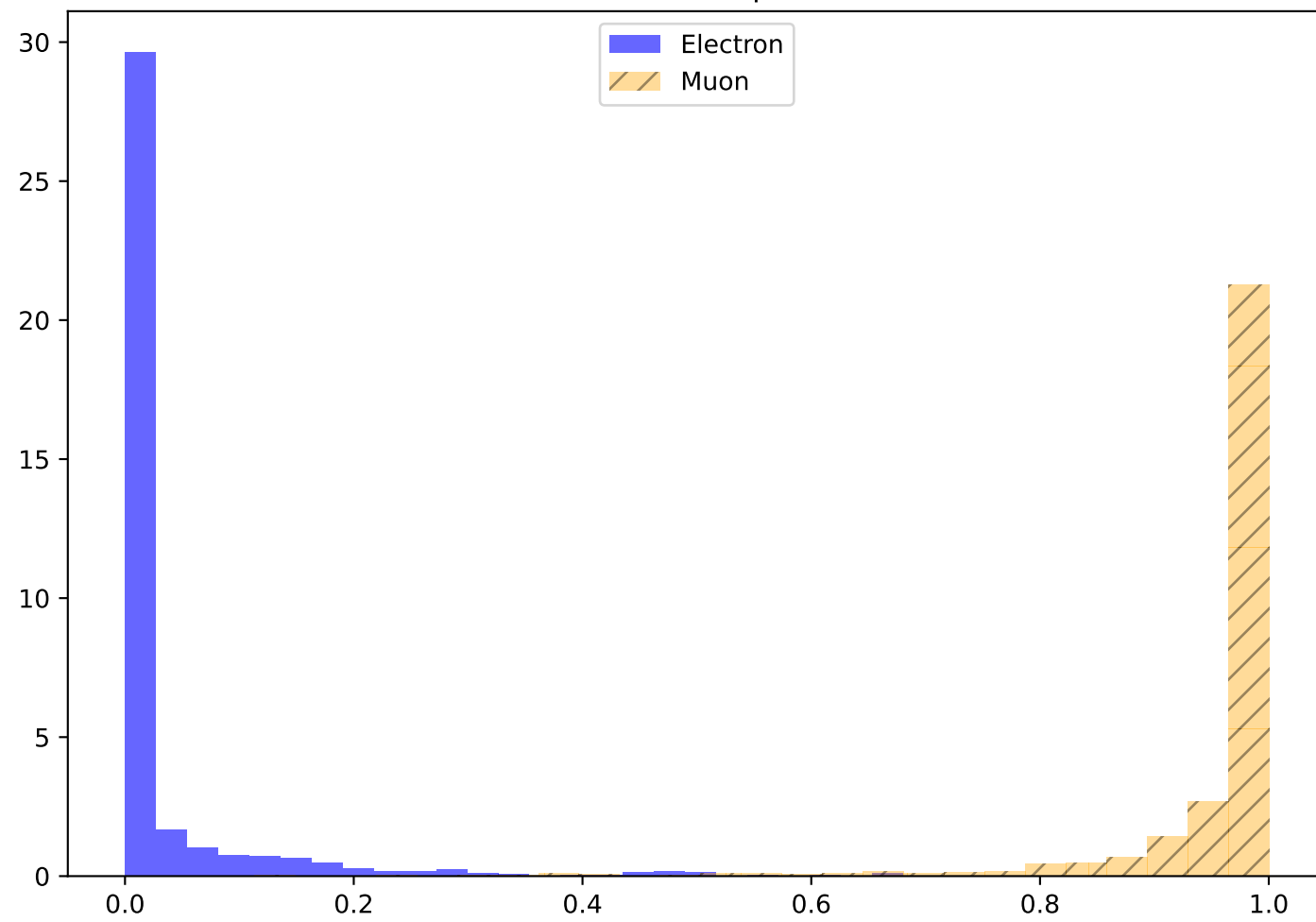
Confusion Matrix



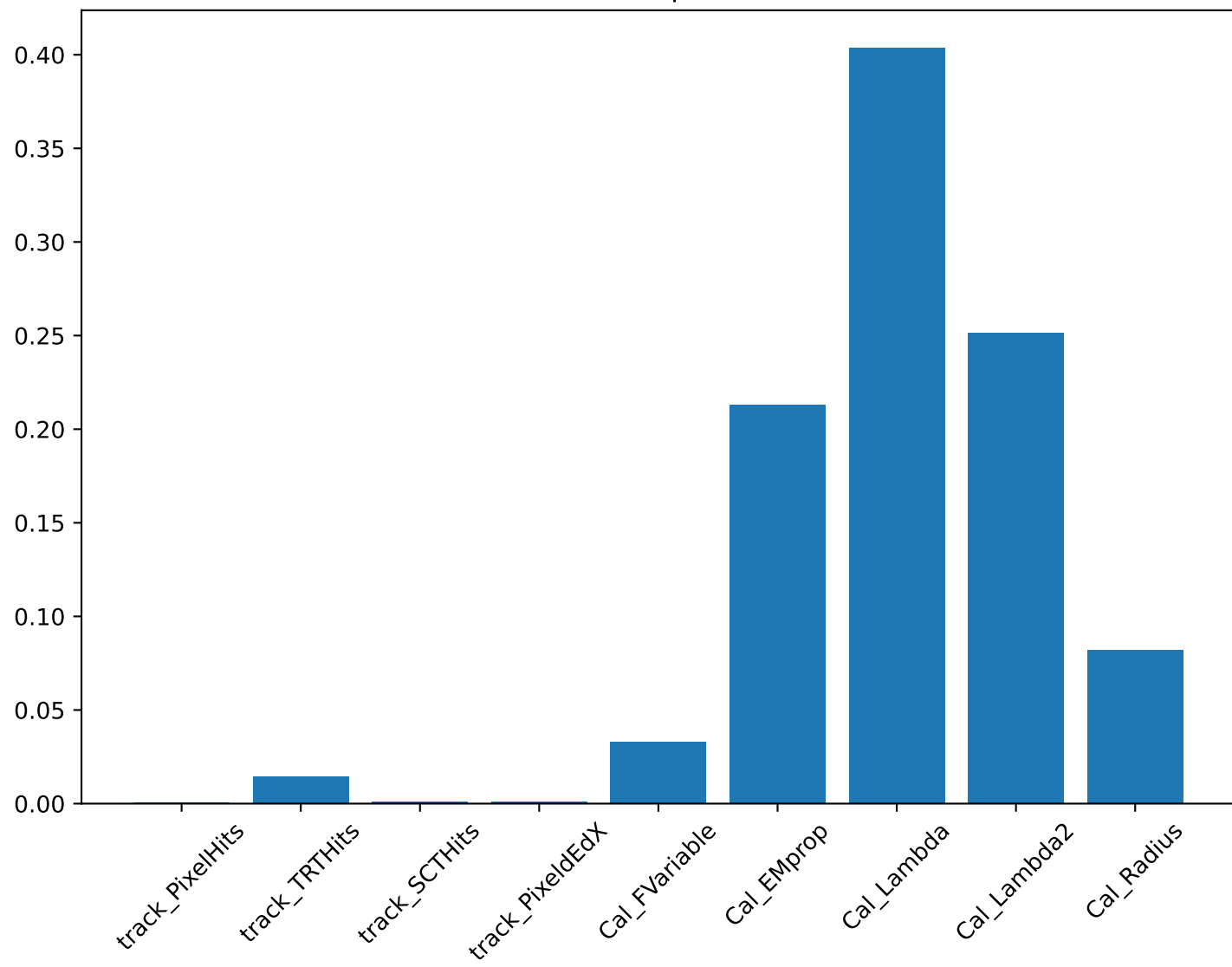
ROC Curve



Classifier Response



Feature Importance



Summary (TEST DATA ONLY)

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	1.00	1.00	1.00	4872
Muon	1.00	1.00	1.00	8735
accuracy			1.00	13607
macro avg	1.00	1.00	1.00	13607
weighted avg	1.00	1.00	1.00	13607

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	1.00	1.00	1.00	2874
Muon	1.00	1.00	1.00	4971
accuracy			1.00	7845
macro avg	1.00	1.00	1.00	7845
weighted avg	1.00	1.00	1.00	7845

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.99	0.99	952
Muon	1.00	1.00	1.00	2667
accuracy			1.00	3619
macro avg	0.99	0.99	0.99	3619
weighted avg	1.00	1.00	1.00	3619

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.99	0.99	0.99	1046
Muon	0.99	0.99	0.99	1097
accuracy			0.99	2143
macro avg	0.99	0.99	0.99	2143
weighted avg	0.99	0.99	0.99	2143